

SOFTWARE ENGINEERING

LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

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1. Lab Objective

The objective of this lab was to use Jira to convert the previously identified functional requirements into Agile backlog items, prioritize and estimate them using Story Points, simulate sprints, and analyse progress using the Sprint Burndown Chart.

2. Jira Project Summary

The Jira workspace used for the simulation was the Software Development space (SWDEV). The main Epic and its child work items were organized as follows:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow donors to post surplus food packages with perishable expiry countdown timers and dietary tags.	High	Pass: Shelter reserves batch and receives pickup verification code; Fail: Expired food batch remains active on claim board.	Allows donors to make surplus food available to shelters before it expires.
FR-002	Functional	The system shall allow verified shelters to reserve a pickup window for an available food batch before its expiry.	High	When a verified shelter selects an available food batch and chooses an available pickup window before expiry, the system records the reservation and displays a confirmation.	Enables shelters to claim available surplus food and arrange collection before expiry.
FR-003	Functional	The system shall allow verified shelters to view available surplus food batches based on location and expiry time	High	When a verified shelter searches for available food, the system displays eligible batches with their location and expiry information.	Helps shelters identify suitable food batches before they expire.
FR-004	Functional	The system shall notify verified shelters within a 5 km radius when high-quality perishable food is listed.	High	When a qualifying food batch is posted, verified shelters within 5 km receive a notification containing the available batch details.	Helps nearby shelters respond quickly to perishable food availability
FR-005	Functional	The system shall provide a pickup verification code for a reserved food batch.	High	After a shelter successfully reserves a food batch, the system generates and displays a unique pickup verification code.	Provides a way to verify the authorized pickup of the reserved food batch.

NFR-001	Nonfunctional	The platform shall dispatch instant push notifications to shelters within a 5 km radius when high-quality perishable food is listed.	High	Pass: Benchmarking tests confirm the target notification latency and security standards under simulated peak load.	Ensures nearby shelters are informed quickly enough to act before perishable food expires.
NFR-002	Nonfunctional	The platform shall prevent unauthorized users from accessing donor and shelter account information.	High	Unauthorized access attempts to protected account information are denied in all security test cases	Protects sensitive information belonging to donors and shelters.

Total estimated effort: 34 Story Points across all seven user stories. Sprint 1 contained the five functional requirements (21 points), while Sprint 2 contained the two non-functional requirements (13 points).

3. Agile Activities Completed

- **Backlog creation and prioritization:** Created the Epic and seven child stories in Jira. The functional requirements and non-functional requirements were prioritized as High, with the Epic retained at Medium.
- **Story Point estimation:** Assigned Fibonacci-style Story Point estimates of 3, 5 and 8 based on relative effort, complexity, and uncertainty.
- **Sprint planning and execution:** Planned Sprint 1 with the five functional requirements (21 points), then progressed the work items through To Do → In Progress → Done. Sprint 2 was then used for the two NFR items (13 points), which were also completed.
- **Progress monitoring:** Used Jira's Sprint Burndown Chart with Story points as the estimation field to observe remaining work and completion.

4. Jira Evidence

The following screenshots document the required Jira deliverables. The screenshots are placed with consistent sizing, spacing, captions, and page breaks for clear evaluation.

Figure 1

Spaces / Software Development / SWDEV-1

Food Bank Surplus Redistribution Platform

▼ Description

The Food Bank Surplus Redistribution Platform is a food rescue platform where restaurants and grocery stores can post surplus edible food batches with expiry information, while verified local shelters can search, receive notifications, and reserve pickup windows before the food expires. The platform supports secure user access and pickup verification to ensure safe and efficient redistribution of surplus food

▼ Attachments

Add attachment Create project plan

▼ Child work items

0% Done

Work	Priority	Assignee	Status
SWDEV-2 FR-001 – Post Surplus Food Batch	High	Unassigned	To Do ▼
SWDEV-3 FR-002 – Reserve Food Pickup Window	High	Unassigned	To Do ▼
SWDEV-4 FR-003 – View Available Food Batches	High	Unassigned	To Do ▼
SWDEV-5 FR-004 – Notify Nearby Shelters	High	Unassigned	To Do ▼
SWDEV-6 FR-005 – Generate Pickup Verification Code	High	Unassigned	To Do ▼
SWDEV-7 NFR-001 – Fast and Secure Notifications	High	Unassigned	To Do ▼
SWDEV-8 NFR-002 – Secure User Account Access	High	Unassigned	To Do ▼

Linked work items

Add linked work item

Add a comment...

Figure 1. Jira Epic and child work items showing the complete backlog structure and priorities.

Figure 2

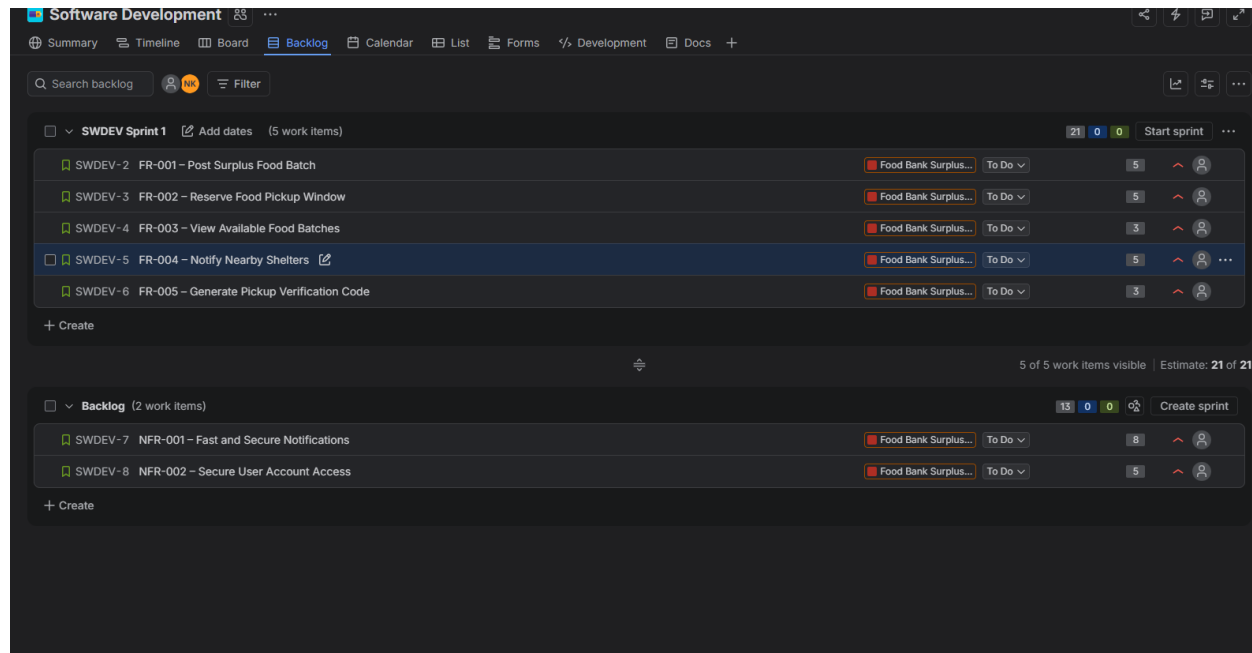


Figure 2. Jira Backlog with Sprint 1 planned and Story Point estimates: 21 points for Sprint 1 and 13 points remaining in the backlog.

Figure 3

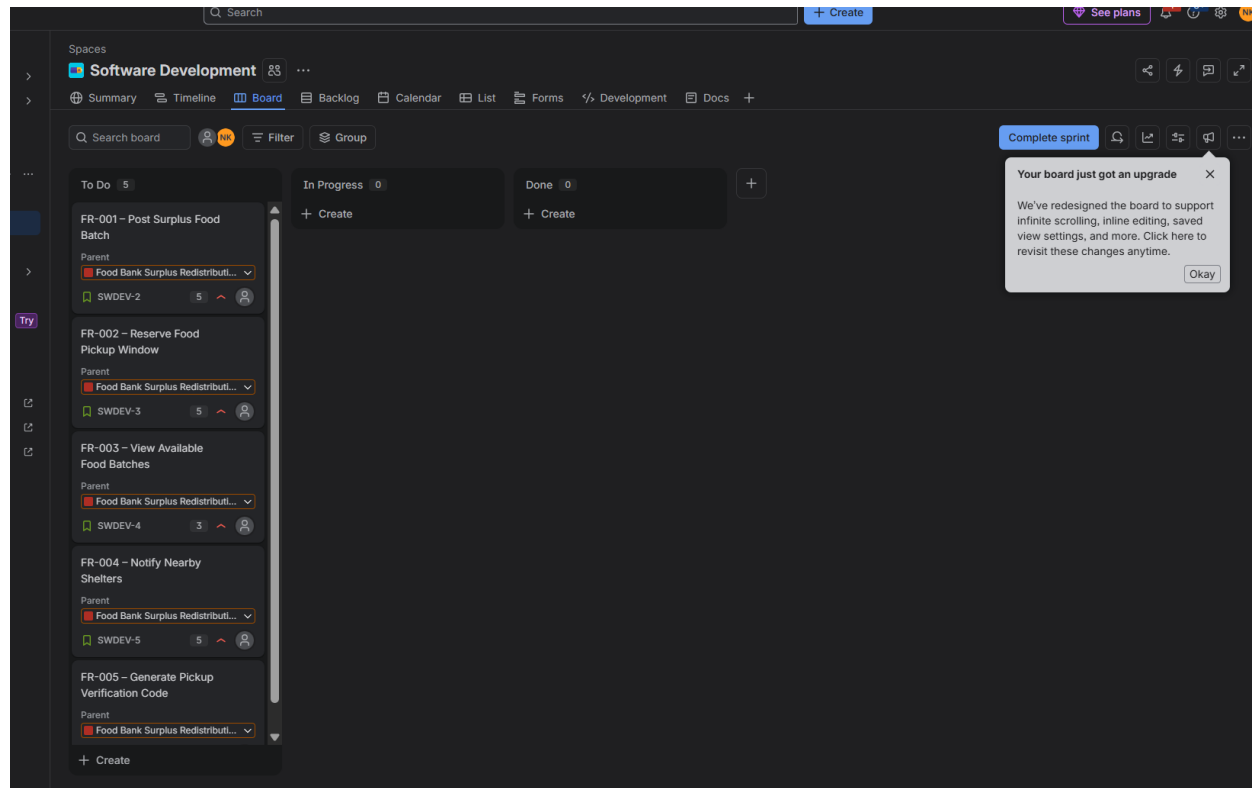


Figure 3. Jira Active Sprint board showing the functional requirements progressing through To Do, In Progress and Done.

Figure 4

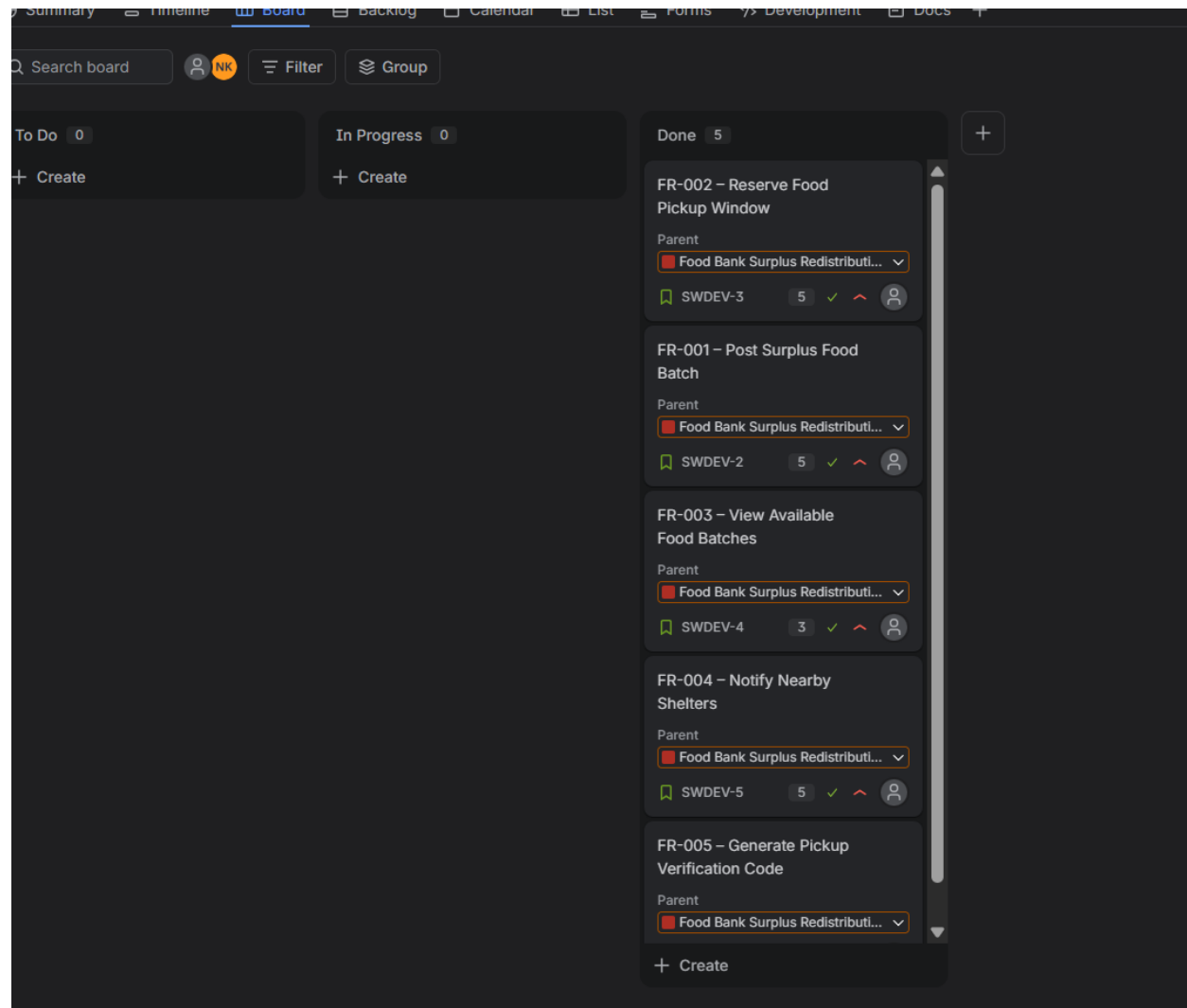


Figure 4. Jira board after the functional sprint work was completed, with all five Sprint 1 items in Done.

Figure 5

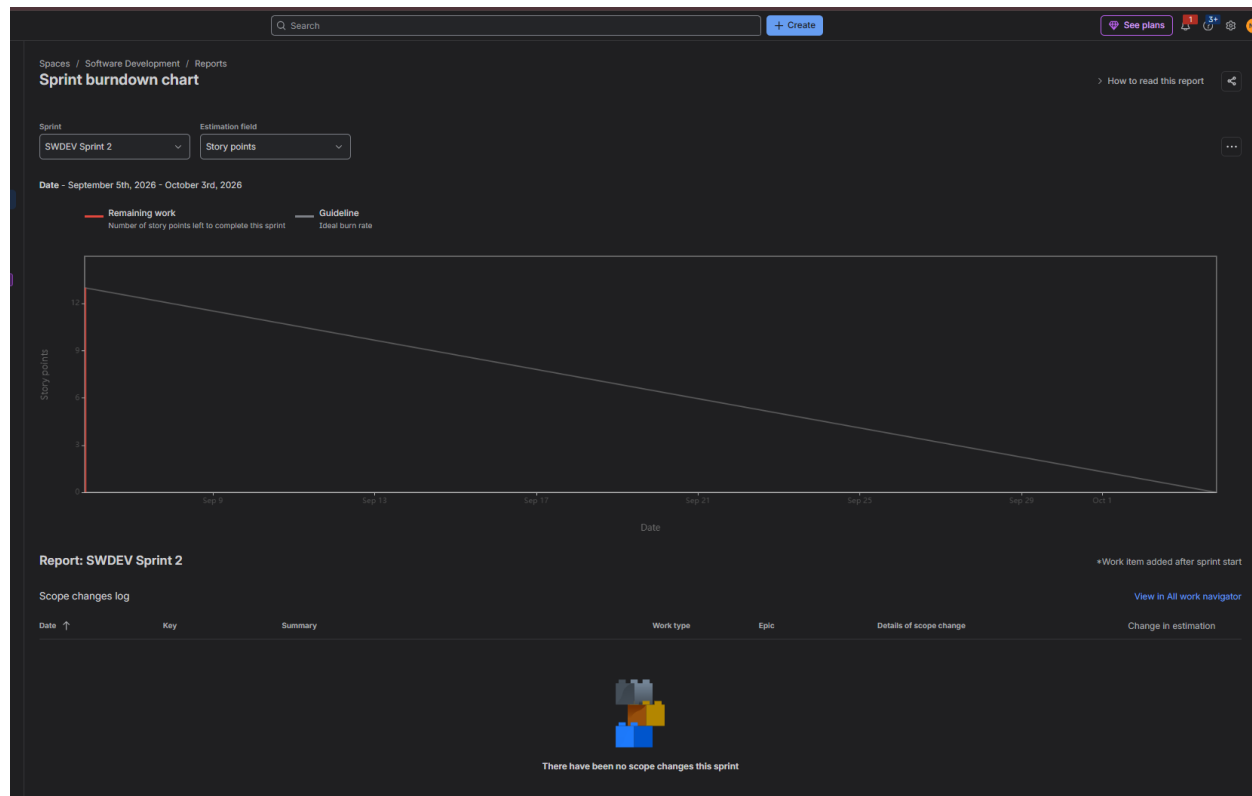


Figure 5. Jira Sprint Burndown Chart for SWDEV Sprint 2 using Story points, showing the sprint burndown and no scope changes.

5. Reflection

1. Did your estimations reflect the actual effort?

Yes, the estimates were reasonable for the simulated Jira exercise. All planned work items were completed within the two sprint simulations, suggesting that the relative estimates were workable for the selected scope. However, because this was a simulation and actual effort in hours was not recorded, the Story Points should be treated as relative estimates rather than a precise measure of time.

2. Was your backlog well-prioritized?

Yes. The functional requirements were treated as the first sprint scope, while the non-functional requirements were handled in the second sprint. The Jira backlog also recorded priorities, with the individual requirements marked High, which kept the important work clearly visible and structured.

3. How did your simulated sprint align with your plan?

The execution aligned well with the plan. Sprint 1 contained the five functional requirements for a total of 21 Story Points, and the items were progressed across the workflow until all five were Done. Sprint 2 contained the two NFR items for 13 Story Points, and both were completed as planned.

4. What insights did the burndown chart give about your team's capacity?

The Sprint 2 burndown showed the planned 13 Story Points reducing to zero remaining work by the end of the sprint, with no scope changes recorded. This indicates that the planned Sprint 2 workload was achievable in the simulation. It also provides a useful reference point for future sprint planning, although one simulated sprint is not enough to establish a reliable long-term team velocity.

6. Conclusion

The Jira Agile simulation was completed successfully. The backlog was structured under a common Epic, the user stories were prioritized and estimated using Story Points, two sprints were executed, and the Sprint Burndown Chart was used to analyse progress. The completed Jira evidence and reflection together document the required Lab 2 activities.